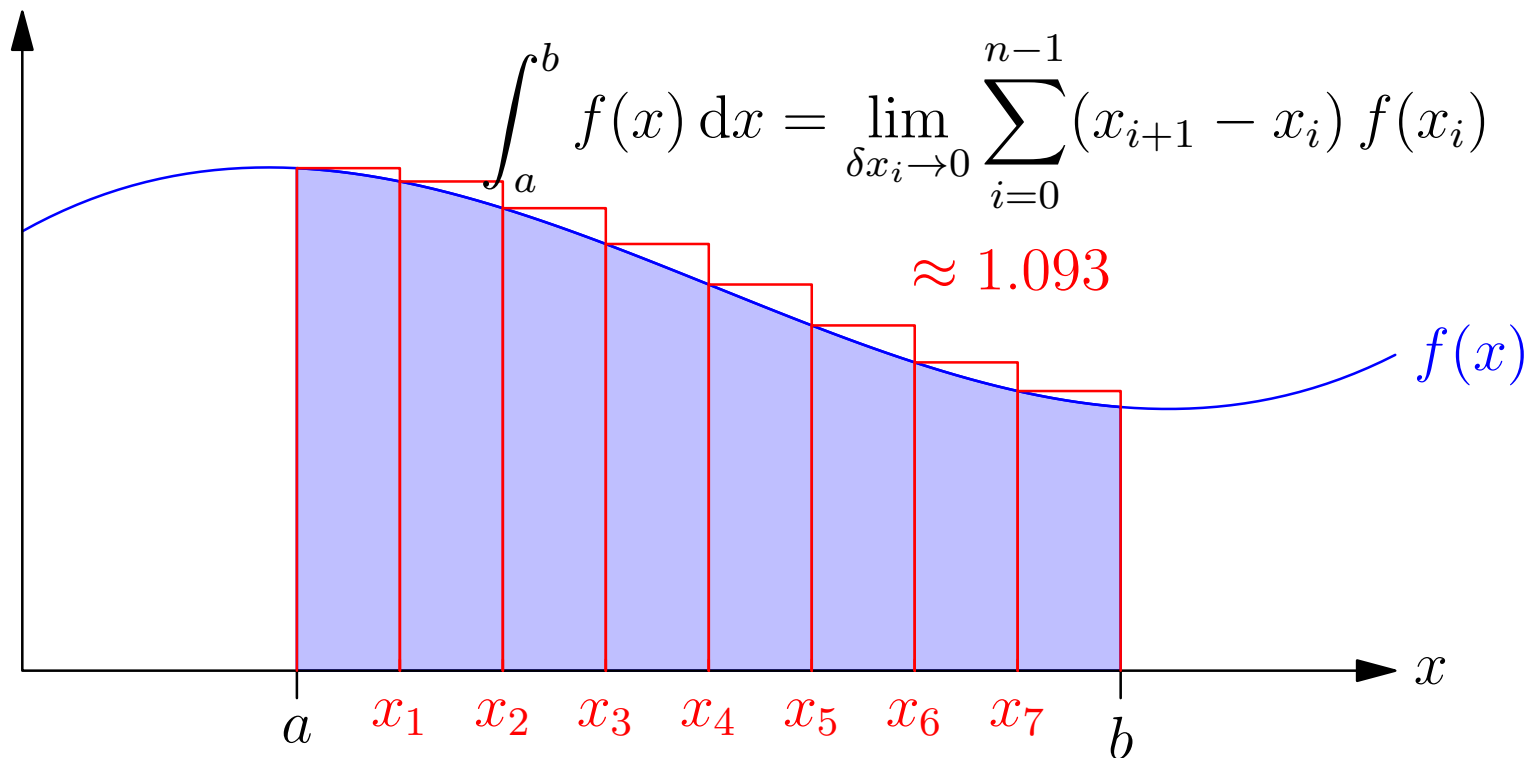


# Advanced Machine Learning

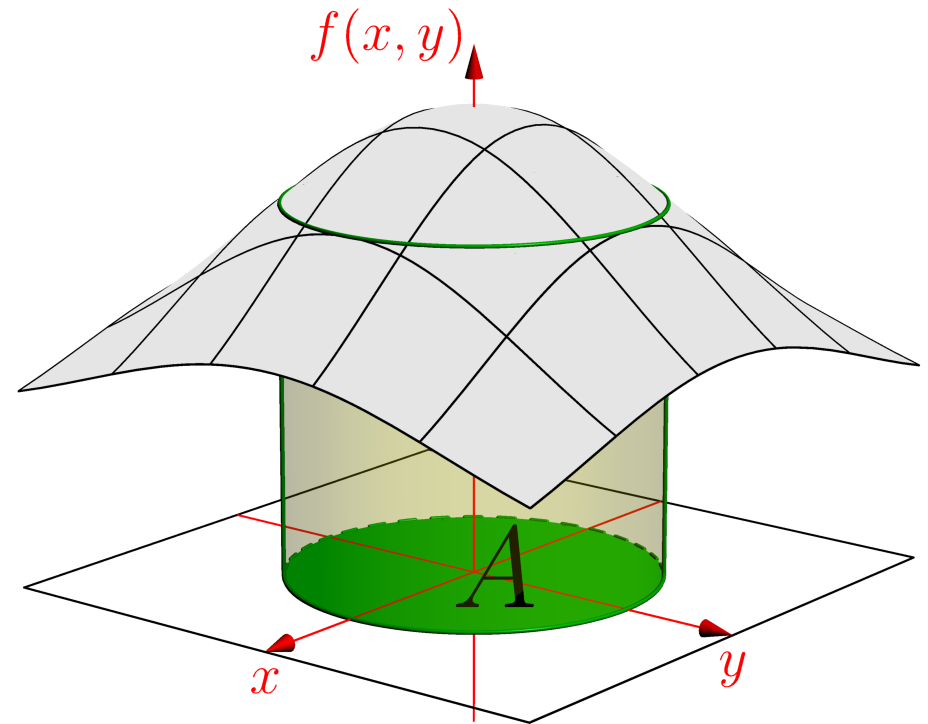
## *Integral Calculus*



*Riemann Integration, integration by parts, gaussian integrals*

# Outline

1. **Defining Integrals**
2. Doing Integrals
3. Gaussian Integrals

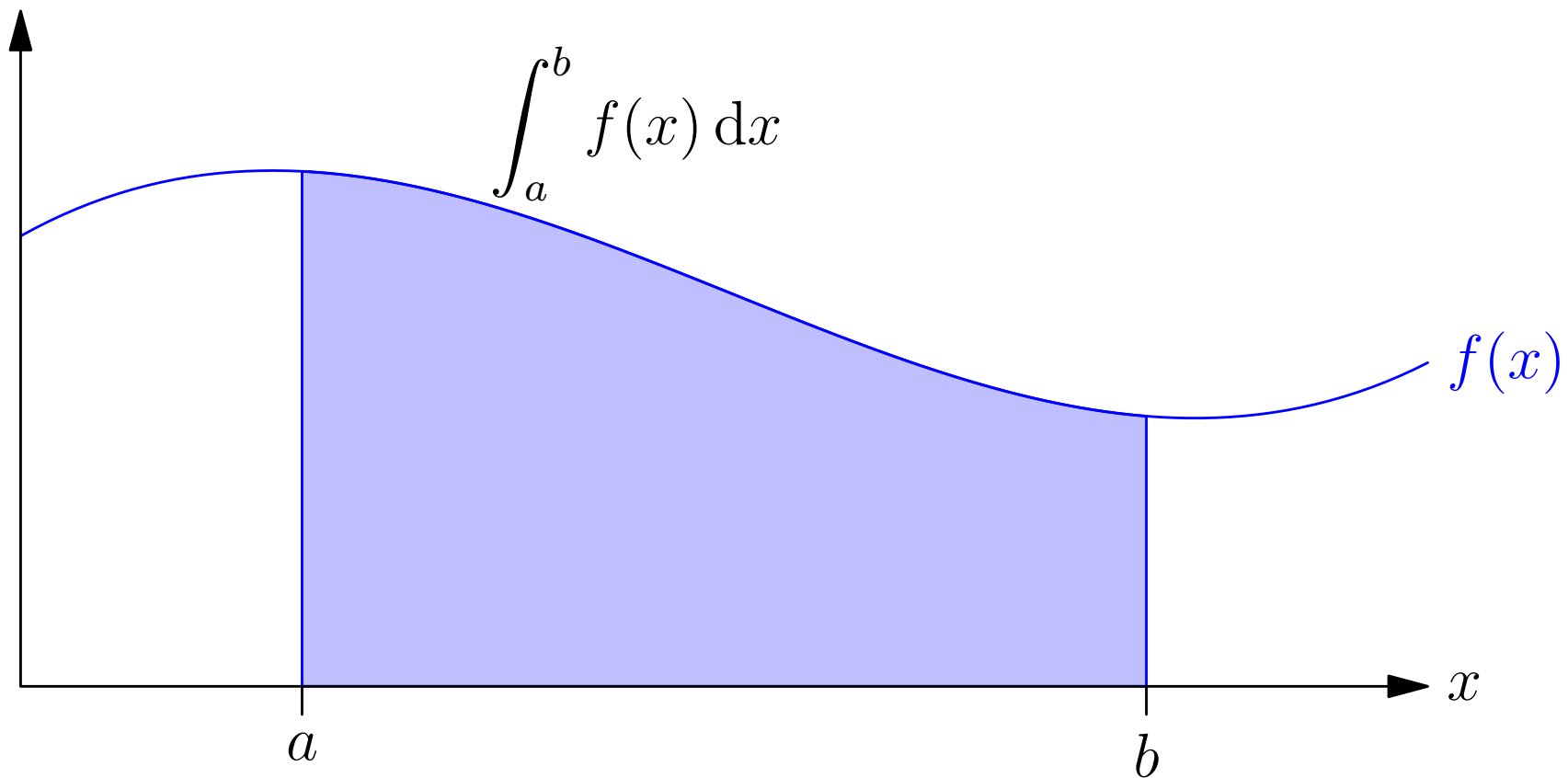


# Riemann Integral

- Integrals represent area beneath a curve

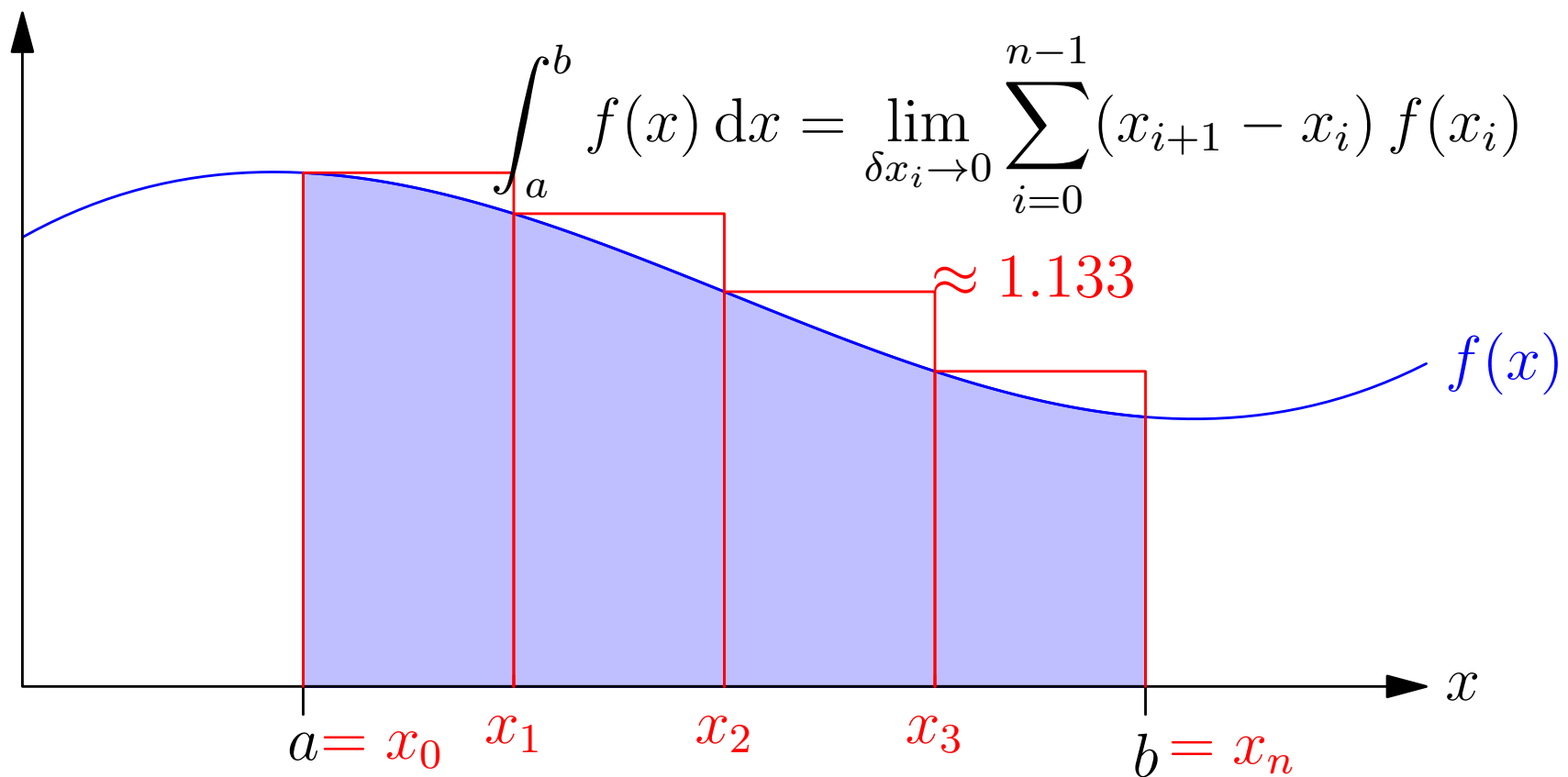
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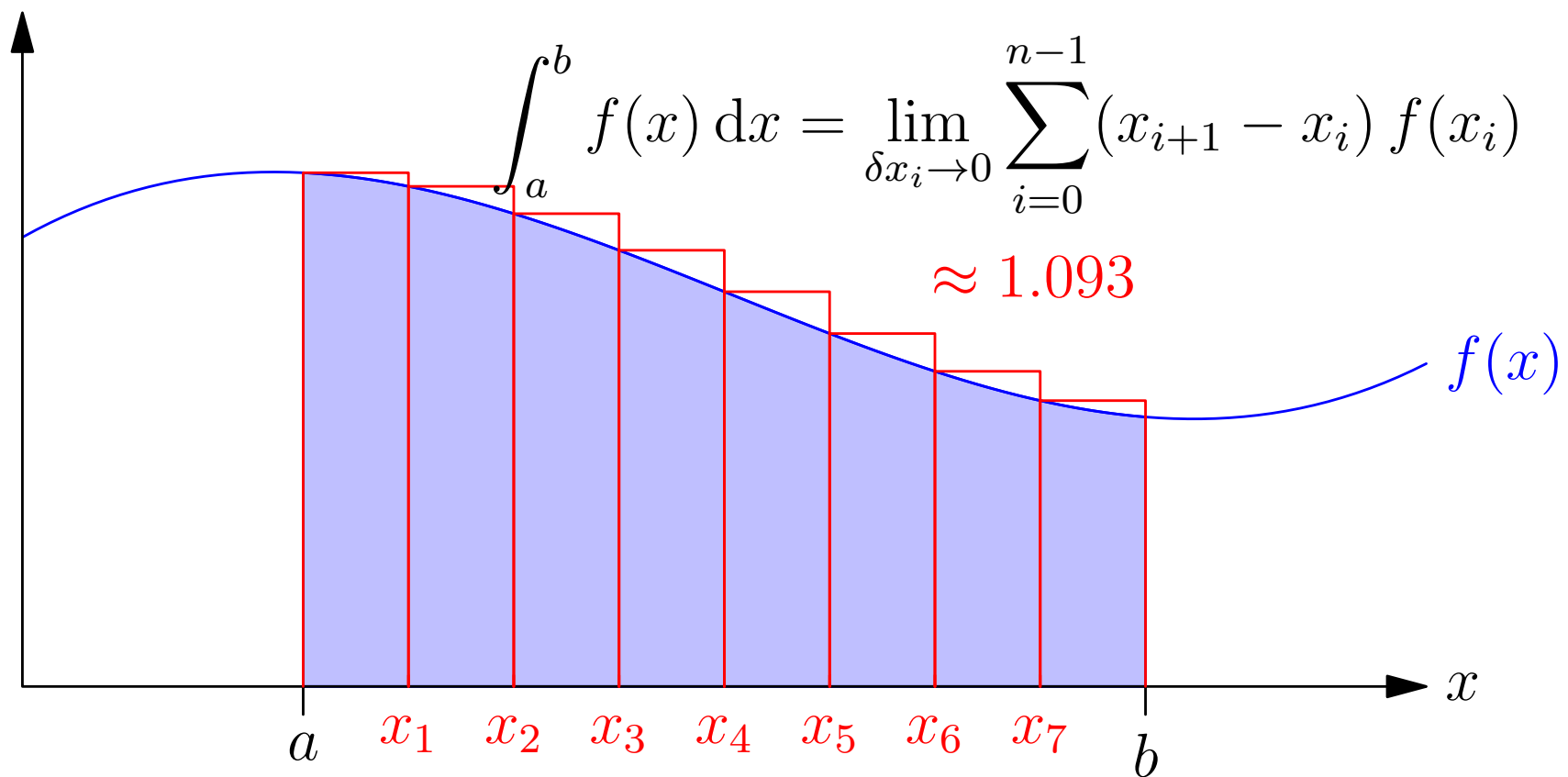
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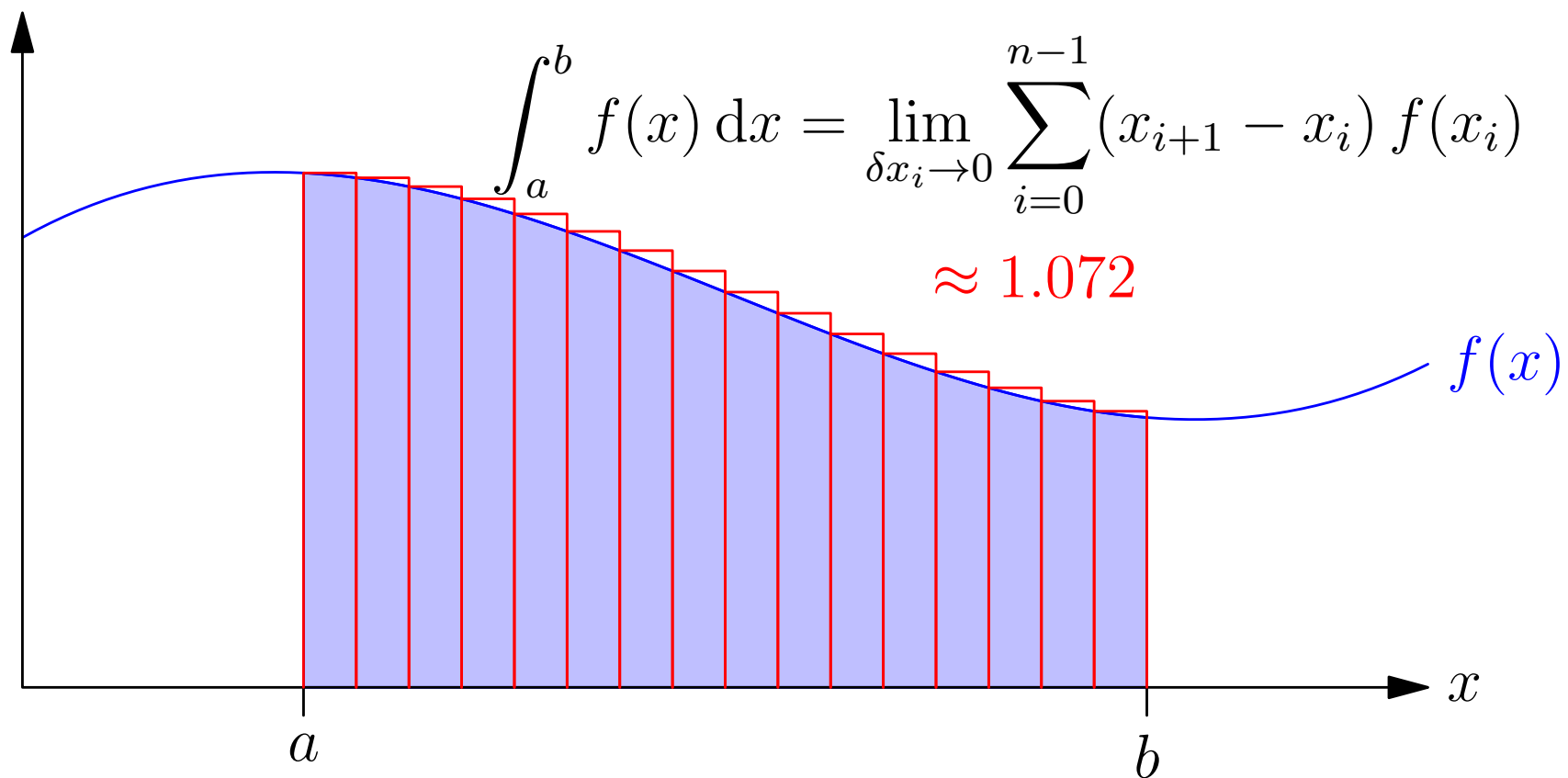
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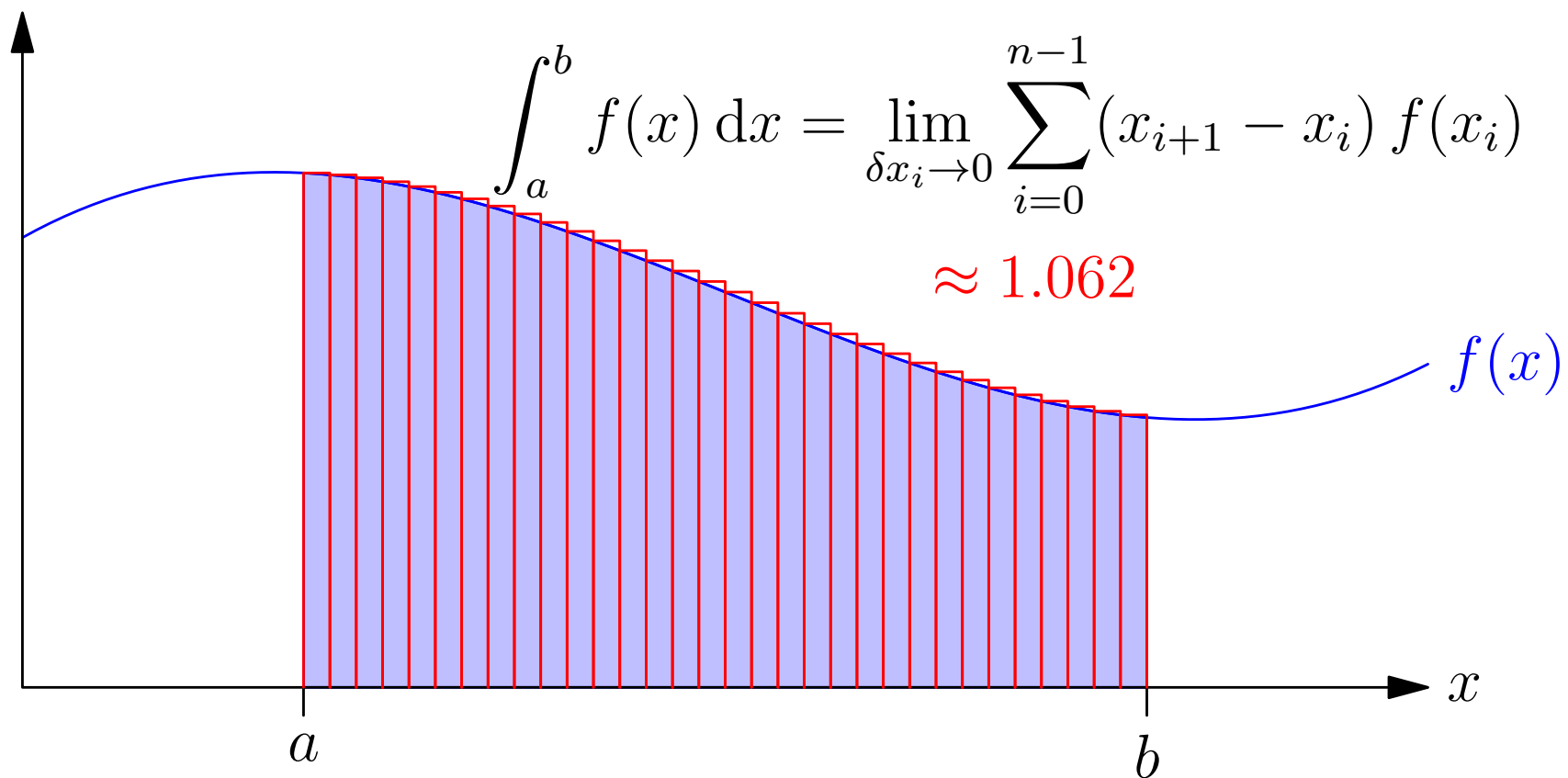
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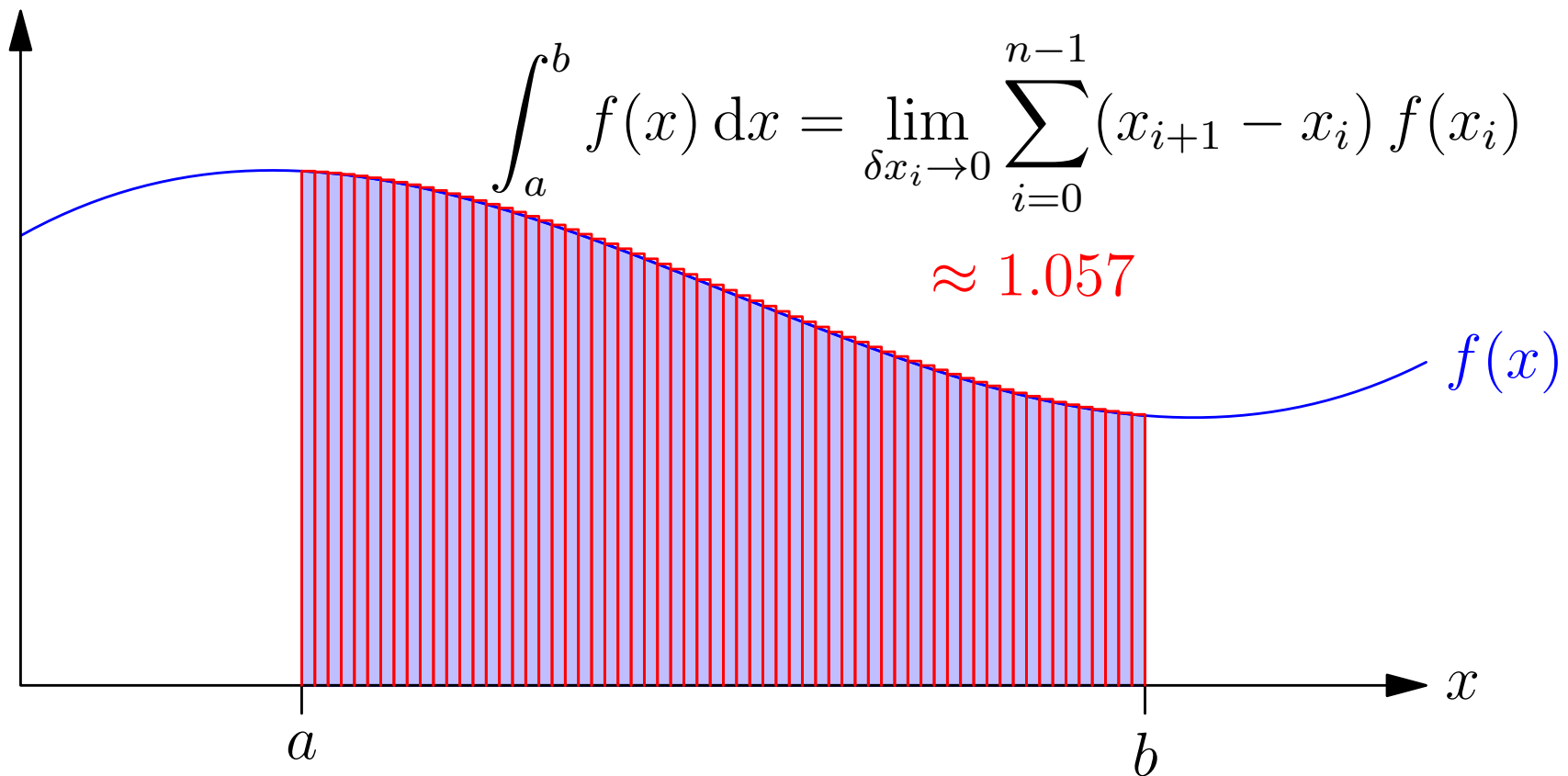
# Riemann Integral

- Integrals represent area beneath a curve



# Riemann Integral

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# Linearity of Integration

- Integration is a linear operator

$$\int_a^b (r f(x) + s g(x)) dx = \lim_{\delta x_i \rightarrow 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) (r f(x_i) + s g(x_i))$$

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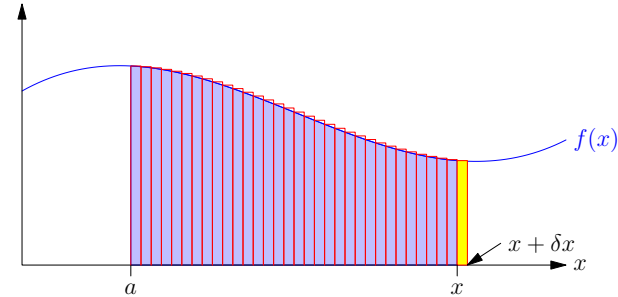
# Fundamental Law of Calculus

- Let

$$I(a, x) = \int_a^x f(z) dz = \lim_{\delta z_i \rightarrow 0} \sum_{i=0}^{n-1} (z_{i+1} - z_i) f(z_i)$$

- Now for small  $\delta x$

$$I(a, x + \delta x) = \int_a^{x + \delta x} f(z) dz$$



- Thus

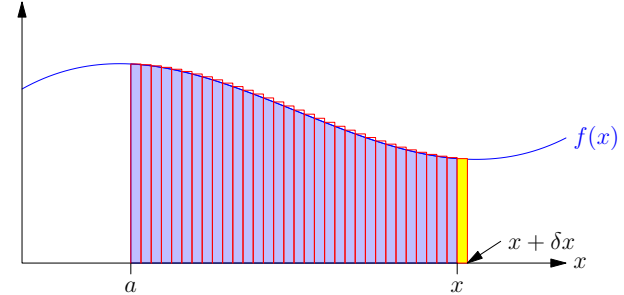
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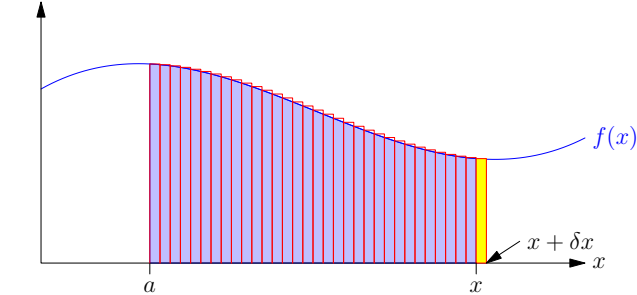
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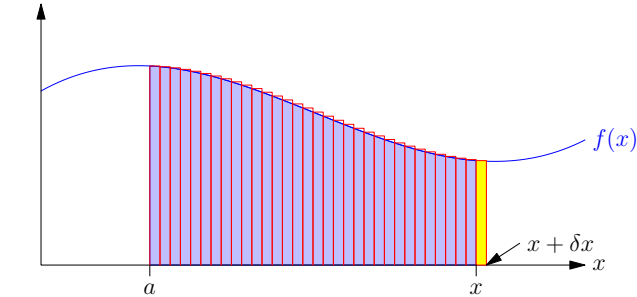
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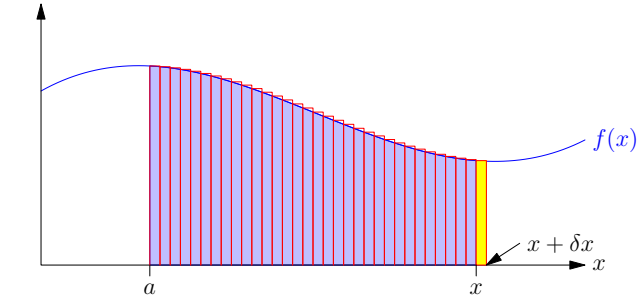
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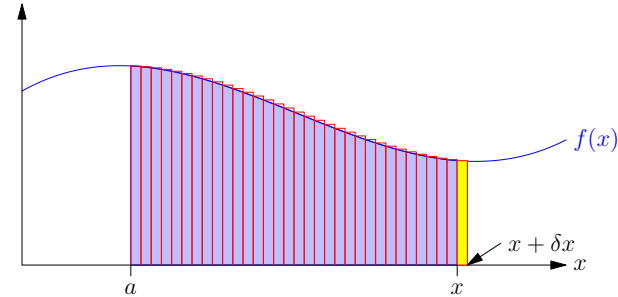
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$$\frac{dI(a, x)}{dx} = \lim_{\delta x \rightarrow 0} \frac{I(x + \delta x) - I(x)}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{\delta x f(x)}{\delta x} = f(x)$$

# The Other Way Around

- Consider

$$\int_a^b \frac{df(x)}{dx} dx = \int_a^b \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} dx$$

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$$\begin{aligned}\int_a^b \frac{df(x)}{dx} dx &= \int_a^b \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} dx \\ &= \lim_{x_{i+1} - x_i \rightarrow 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}\end{aligned}$$

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- We can think of integration as an **anti-derivative**—it undoes differentiation

# Indefinite Integrals

- So far we have considered **definite integrals** where we integrate between two points ( $a$  and  $b$ )
- However, when think about integration as an anti-derivative, it is useful to think of a function  $F(x) = \int f(x)dx$
- So that  $F'(x) = f(x)$
- However the function  $F(x)$ ,  $F(x) + 1$ ,  $F(x) + \pi$ , etc. all have the same derivative so  $F(x)$  is only defined up to an additive constant
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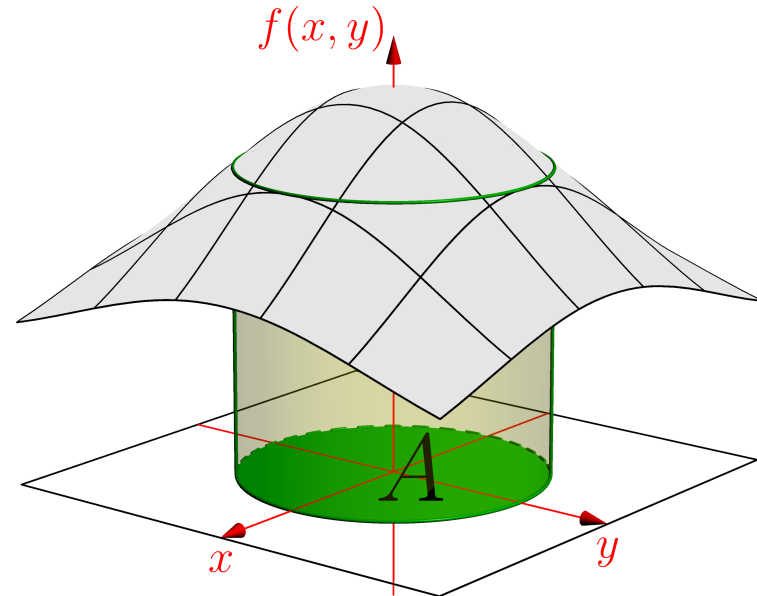
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# Multiple Integrals

- For functions involving many independent variables (e.g.  $f(x,y)$ ,  $f(x,y,z)$ ,  $f(\mathbf{x})$ ) we can integrate over multiple dimensions
- For example

$$\iint_A f(x,y) dx dy$$



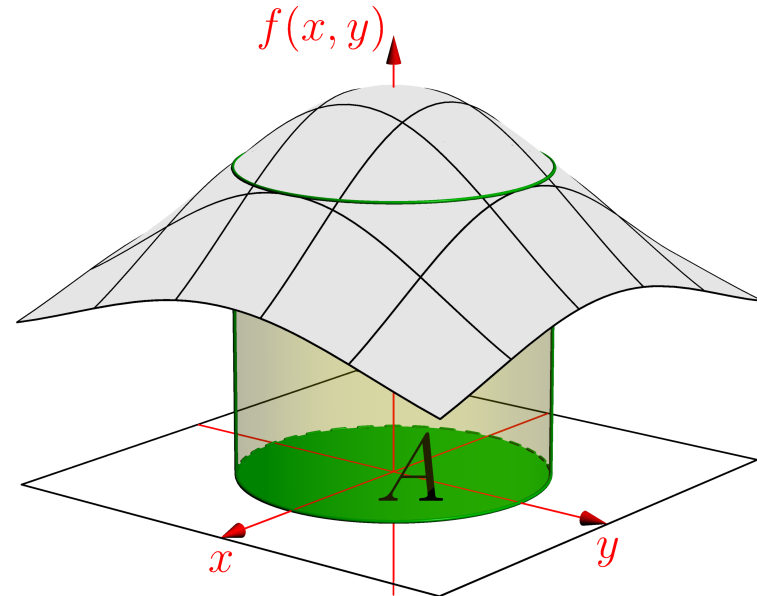
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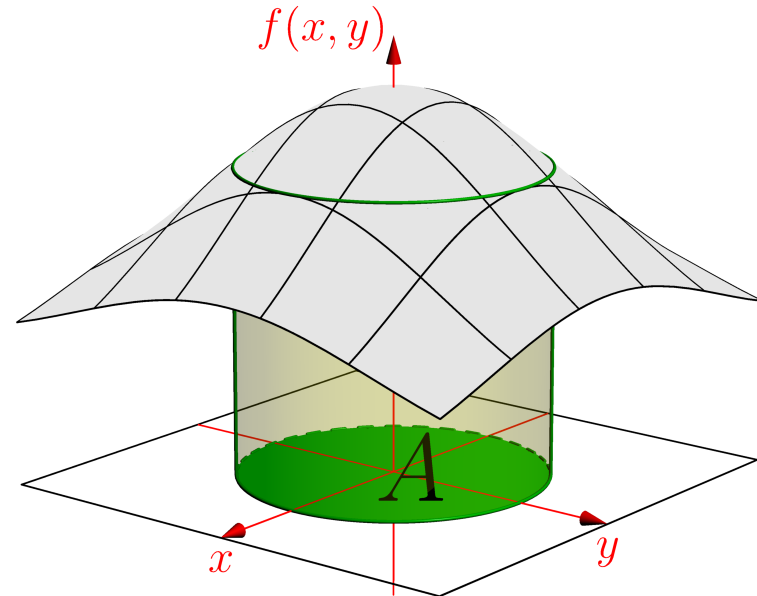
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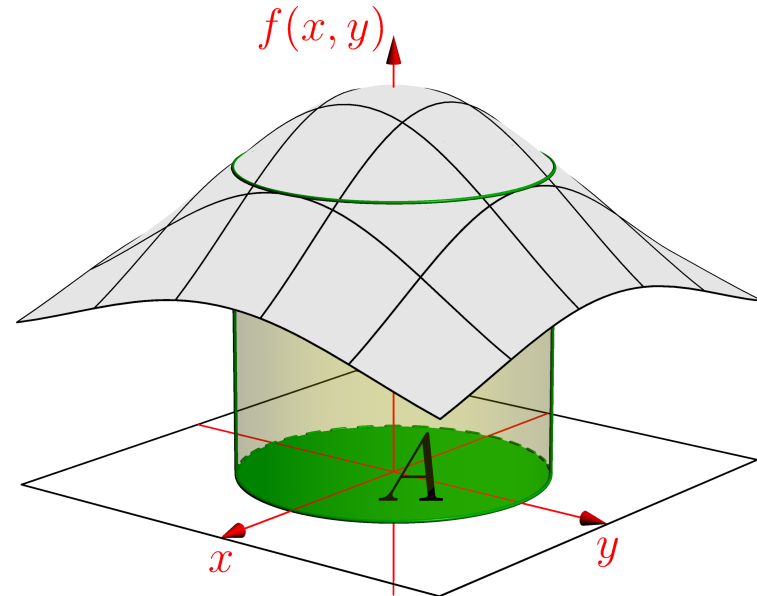
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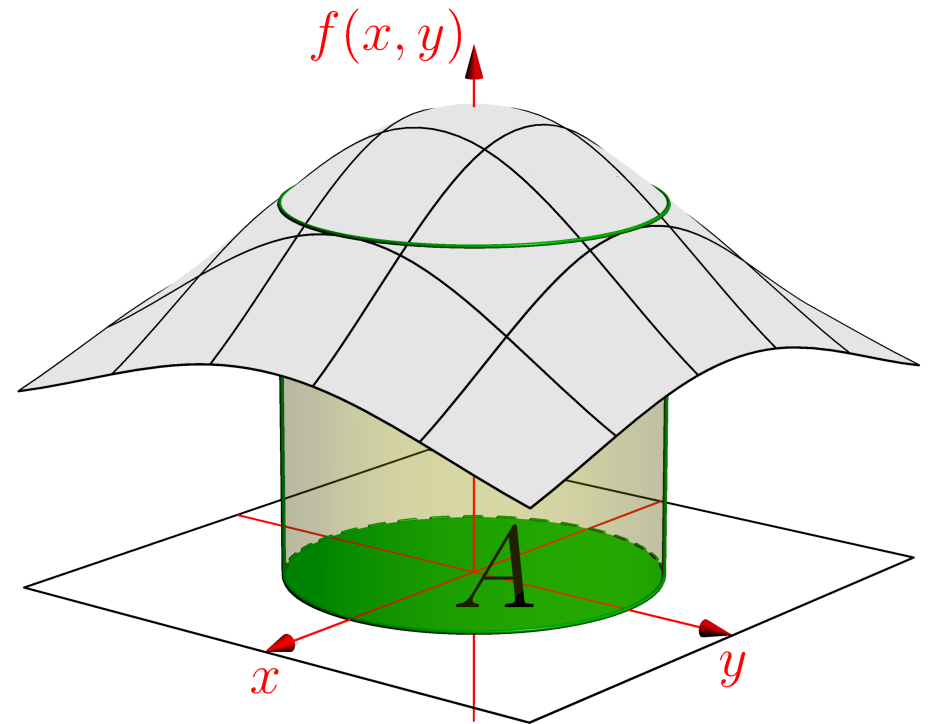


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# Outline

1. Defining Integrals
2. **Doing Integrals**
3. Gaussian Integrals



# Performing Integration

- A key method for performing integrals is through knowledge of the anti-derivative
- If we know  $F'(x) = f(x)$  then  $F(x) + c = \int f(x) dx$
- E.g. we know that  $dx^n/dx = nx^{n-1}$  therefore

$$\int x^{n-1} dx = \frac{1}{n} \int \frac{dx^n}{dx} dx$$

and

$$\int_a^b x^{n-1} dx = \frac{b^n}{n} - \frac{a^n}{n}$$

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$$\int x^{n-1} dx = \frac{1}{n} \int \frac{dx^n}{dx} dx = \frac{x^n}{n} + c$$

and

$$\int_a^b x^{n-1} dx = \frac{b^n}{n} - \frac{a^n}{n}$$

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- We saw, due to the product and chain rules, that we can differentiate almost anything

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whether this is helpful depends on  $f(x)$  and  $g(x)$

# Example of Integration by Parts

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- Now

$$\Pi(n) = n\Pi(n-1) = n(n-1)\Pi(n-2) = n(n-1)(n-2)\dots 1 = n!$$

# Substitution

- We can make a transformation from  $x$  to  $u$

$$\int_a^b f(x) dx = \lim_{\delta x_i \rightarrow 0} \sum_{i=0}^{n-1} f(x_i)(x_{i+1} - x_i)$$

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- We consider  $I(n) = \int_0^{\infty} x^n e^{-x^2/2} dx$
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# Changing Variables in Multidimensional Space

- When changing variables in many dimensions  $\mathbf{x} \rightarrow \mathbf{u}$  the change of variables involves the Jacobian

$$\int f(\mathbf{x}) d\mathbf{x} = \int f(\mathbf{x}(\mathbf{u})) |\det(\mathbf{J})| d\mathbf{u}, \quad \mathbf{J} = \begin{pmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_1}{\partial u_2} & \cdots & \frac{\partial x_1}{\partial u_n} \\ \frac{\partial x_2}{\partial u_1} & \frac{\partial x_2}{\partial u_2} & \cdots & \frac{\partial x_2}{\partial u_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial u_1} & \frac{\partial x_n}{\partial u_2} & \cdots & \frac{\partial x_n}{\partial u_n} \end{pmatrix}$$

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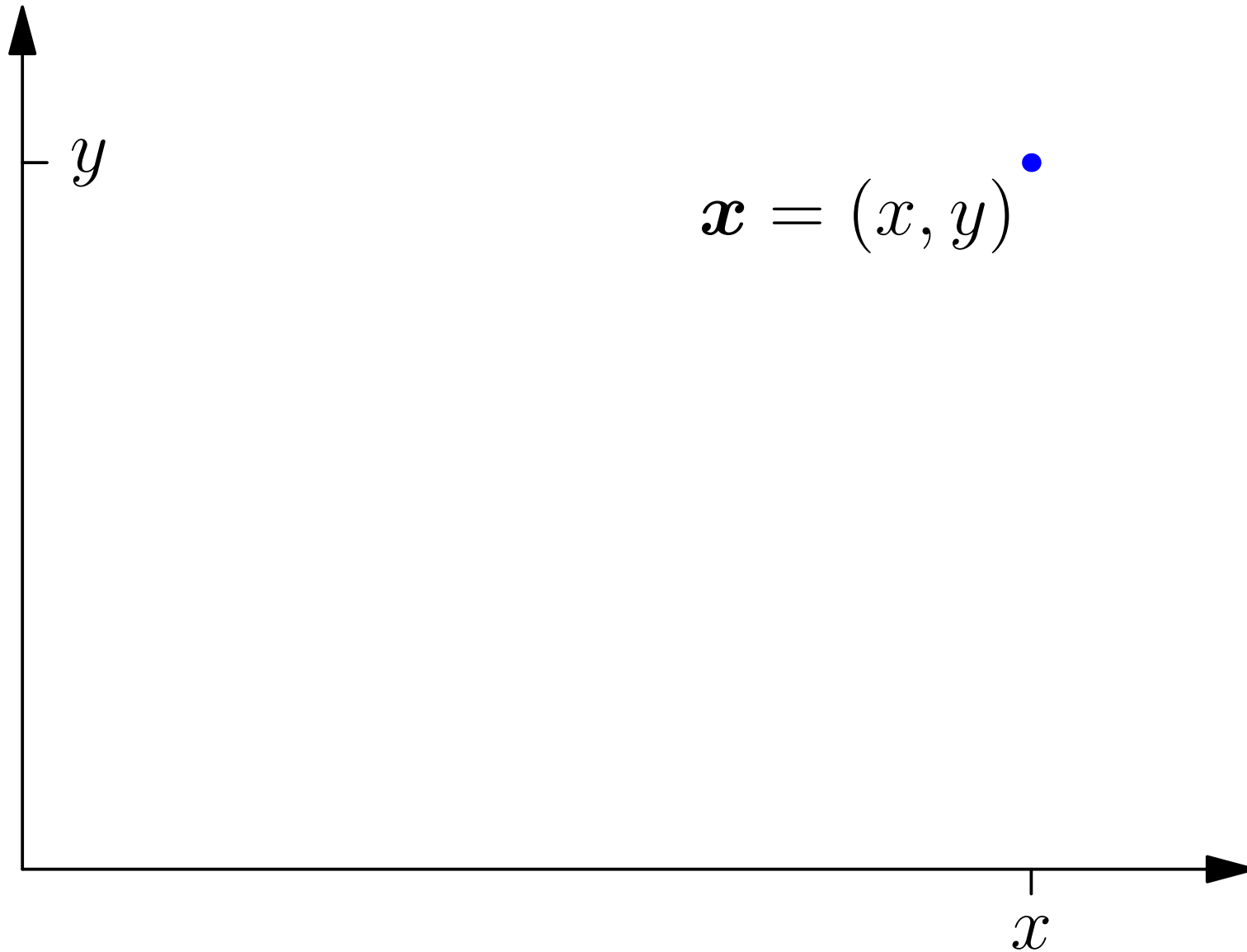
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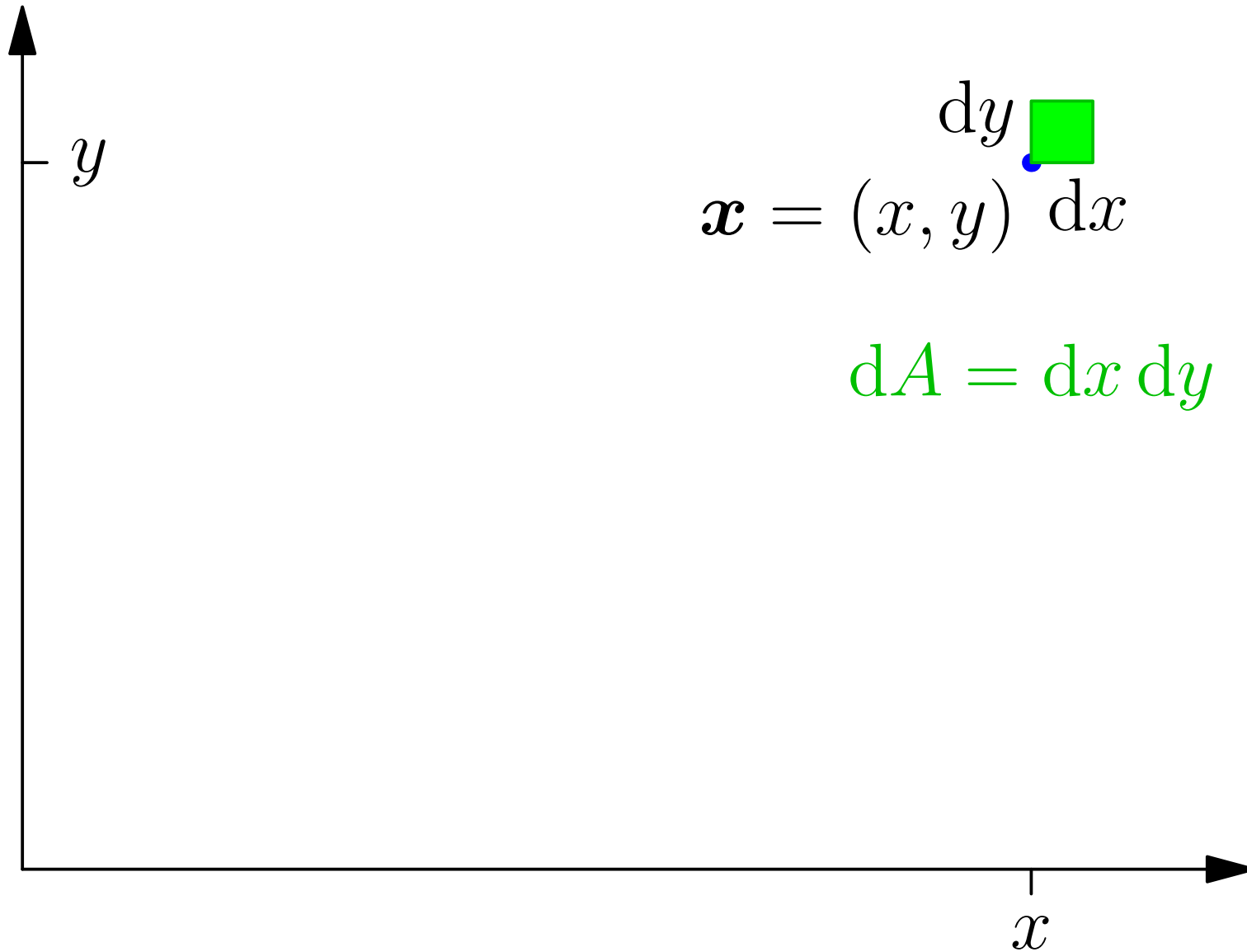
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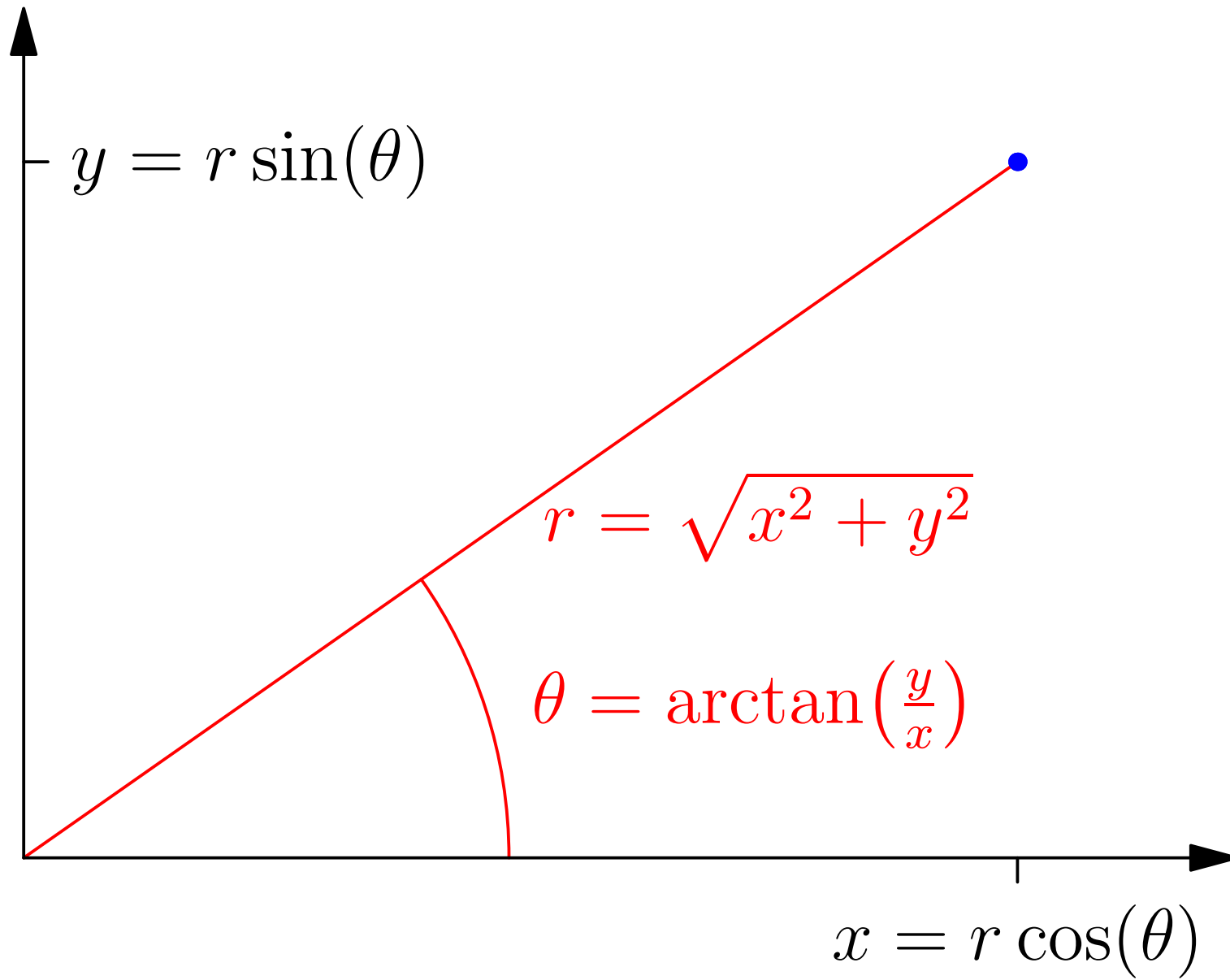
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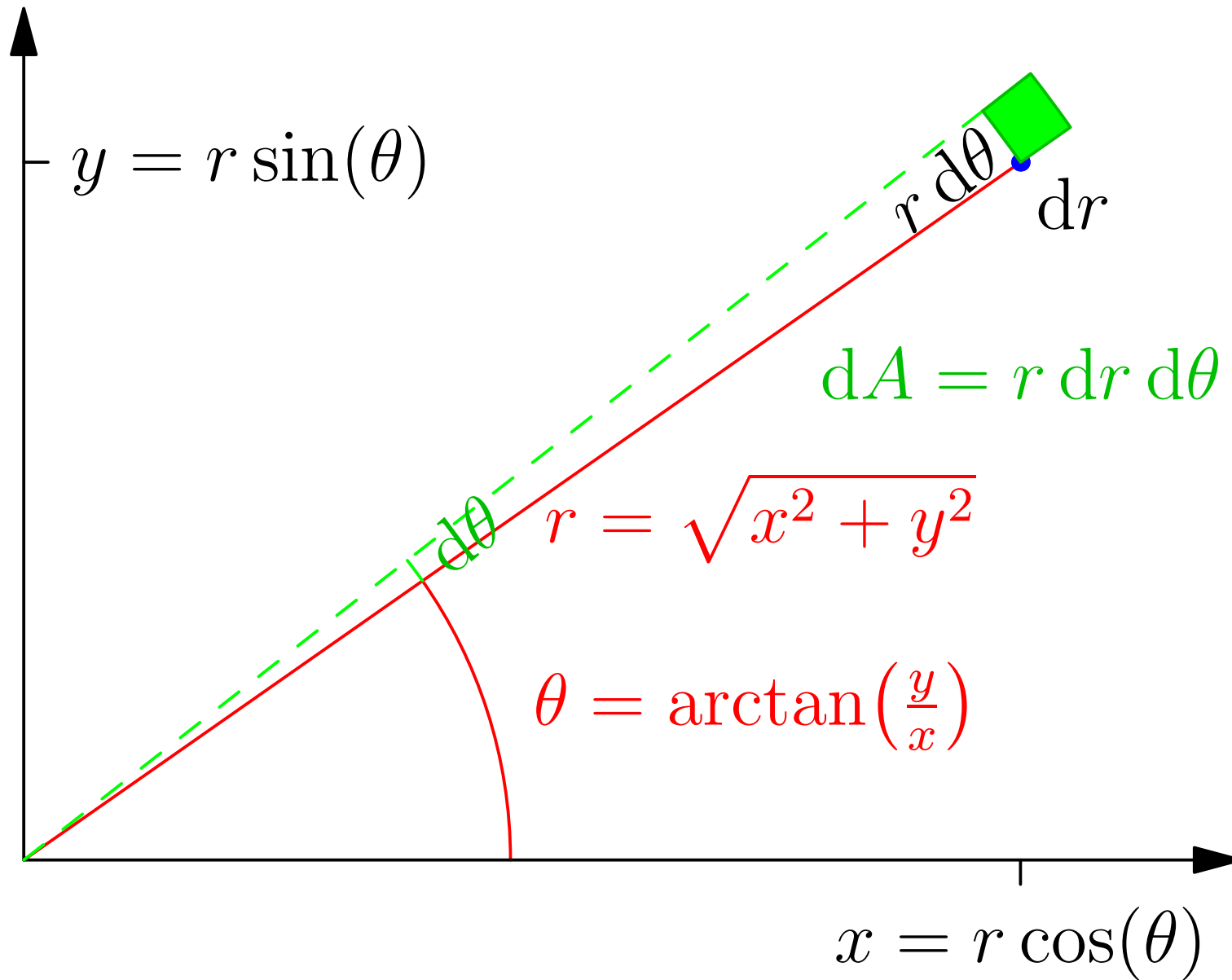
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# Differentiating Through the Integral

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$$M_n = \mathbb{E}[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) dx$$

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# Special Functions

- There are integrals with no known closed form solution
- We saw that  $\Pi(z) = \int_0^{\infty} x^z e^{-x} dx$  satisfies  $\Pi(z) = z\Pi(z-1)$
- For integer  $n$  then  $\Pi(n) = n!$ , but for general  $z$ , the integral  $\Pi(z)$  can't be written in terms of elementary functions
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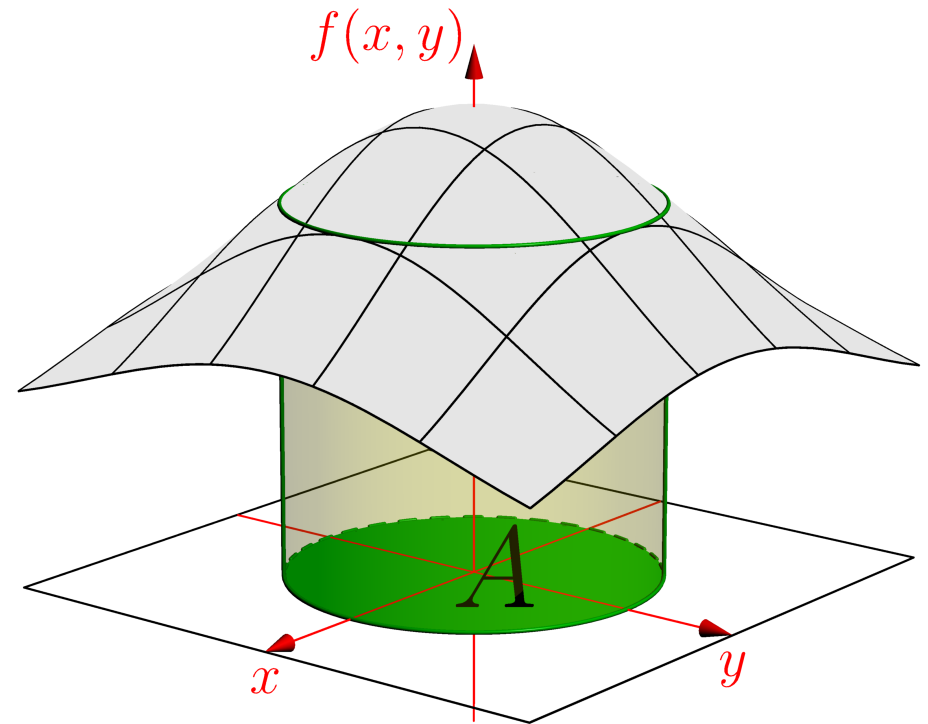
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- Other special function defined by integrals exist (e.g. the Bessel , Aire, hypergeometric, elliptic, error functions, . . . )

# Outline

1. Defining Integrals
2. Doing Integrals
3. **Gaussian Integrals**



# Gaussian Integrals

- Gaussian integrals are integrals involving  $e^{-x^2}$ , e.g.

$$\int_{-\infty}^{\infty} e^{-x^2} dx$$

$$\int_{-\infty}^{\infty} x^4 e^{-ax^2 - bx} dx$$

- They are important in computing integrals with respect to the normal distribution

$$\mathcal{N}(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$$

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- The bad news is that they are quite tricky to do

# The Gaussian Integral

- The integral over a Gaussian is surprisingly difficult

$$I_1 = \int_{-\infty}^{\infty} e^{-x^2/2} dx$$

- There is a nice trick which is to consider

$$I_1^2 = \int_{-\infty}^{\infty} e^{-x^2/2} dx \int_{-\infty}^{\infty} e^{-y^2/2} dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy$$

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- From before

$$I_1^2 = 2\pi \int_0^\infty r e^{-r^2/2} dr$$

- Finally let  $u = r^2/2$  so that  $du/dr = r$  or  $du = r dr$  we get

$$I_1^2 = 2\pi \int_0^\infty e^{-u} du = 2\pi$$

- So that  $I_1 = \sqrt{2\pi}$
- Incidentally,  $I_1 = \sqrt{2}\Pi(-1/2)$  so  $\Pi(-1/2) = \Gamma(1/2) = \sqrt{\pi}$

# Normal Distribution

- We consider

$$I_2 = \int_{-\infty}^{\infty} e^{-(x-\mu)^2/(2\sigma^2)} dx$$

- Making the change of variables  $z = (x - \mu)/\sigma$  so that  $dz = dx/\sigma$  or  $dx = \sigma dz$ . Then

$$I_2 = \sigma \int_{-\infty}^{\infty} e^{-z^2/2} dz = \sigma I_1 = \sqrt{2\pi} \sigma$$

- Note that the *probability density function* (PDF) for a normally distributed random variable is given by

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# Multi-dimensional Gaussians

- Consider

$$I_3 = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} e^{-\frac{1}{2}\|\mathbf{x}\|_2^2} dx_1 \cdots dx_n$$

where  $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$

- Note that  $\|\mathbf{x}\|_2^2 = x_1^2 + x_2^2 + \cdots + x_n^2$  and using  $e^{\sum_i a_i} = \prod_i e^{a_i}$

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# Full Multi-variate Normal

- Consider

$$I_4 = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Xi}^{-1}(\mathbf{x}-\boldsymbol{\mu})} d\mathbf{x}_1 \cdots d\mathbf{x}_n$$

- Let  $\boldsymbol{\Xi}^{-1} = \mathbf{V}\boldsymbol{\Lambda}^{-1}\mathbf{V}^T$  and make the change of variables  $\mathbf{y} = \mathbf{V}^T(\mathbf{x} - \boldsymbol{\mu})$
- The Jacobian  $\mathbf{J}$  has elements (note that  $\mathbf{x} = \mathbf{V}\mathbf{y} + \boldsymbol{\mu}$ )

$$J_{ij} = \frac{\partial x_i}{\partial y_j} = \frac{\partial}{\partial y_j} \left( \sum_{k=1}^n V_{ik} y_k + \mu_i \right) = V_{ij}$$

- So that  $\mathbf{J} = \mathbf{V}$  and consequently  $|\det(\mathbf{J})| = |\det(\mathbf{V})| = 1$  then

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# Determinants

- Using the facts, that  $\Xi = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$  then

$$\det(\Xi) = \det(\mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top)$$

- Recall  $I_4 = \prod_i \sqrt{2\pi\lambda_i}$
- We note for an  $n \times n$  matrix  $\mathbf{M}$  then  $\det(c\mathbf{M}) = c^n \det(\mathbf{M})$  so that

$$I_4 = (2\pi)^{n/2} \sqrt{\det(\Xi)} = \sqrt{\det(2\pi\Xi)}$$

- Finally, we get that for the PDF of a normal to integrate to 1

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \Xi) = \frac{1}{\sqrt{\det(2\pi\Xi)}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Xi^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$

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- Making friends with integration will give you a super-power that not too many people share